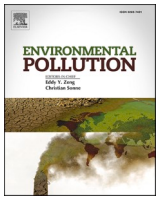
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Water quality forecasting based on data decomposition, fuzzy clustering and deep learning neural network☆

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ABSTRACT

Water quality forecasting can provide useful information for public health protection and support water resources management. In order to forecast water quality more accurately, this paper proposes a novel hybrid model by combining data decomposition, fuzzy C-means clustering and bidirectional gated recurrent unit. Firstly, the original water quality data is decomposed into several subseries by empirical wavelet transform, and then, the decomposed subseries are recombined by fuzzy C-means clustering. Next, for each clustered series, bidirectional gated recurrent unit is applied to develop prediction model. Finally, the forecast result is obtained by the summation of the predictions for the subseries. The proposed forecast model is evaluated by the water quality data of Poyang Lake, China. Results show that the proposed forecast model provides highly accurate forecast result for all of the six water quality data: the average of MAPE of the forecast results for the six water quality datasets is 4.59% for 7 day ahead prediction. Furthermore, our model shows better forecast performance than the other models. Particularly, compared with the single BiGRU model, MAPE decreased by 32.86% in average. Results demonstrate that the proposed forecast model can be used effectively for water quality forecasting.

1. Introduction

Water is one of the most precious natural resources for human beings. In the last decades, with the development of the economy, water consumption in industry and people's lives increased rapidly, and therefore, good water quality has been becoming one of the most important problems that people faced globally (Aldhyani et al., 2020; Ding et al., 2021). In the 21st century, water pollution has always been a serious social, economic and environmental problem that hinders economic development and causes threat to public health (Farzin et al., 2020; Loos et al., 2020). Since water quality is closely connected with economy and people's lives, accurate water quality forecasting is of great significance. Water quality forecasting can be successfully applied to many fields, such as, drinking water supply, irrigation, fishing industry, tourism industry, etc (Peng et al., 2019). Moreover, it can provide useful information for the water resource management to protect water quality (Li et al., 2017).

Given the importance of water quality forecasting, many researches have been conducted to develop water quality forecast model (Vincon-Leite and Casenave, 2019). The researches on this issue are performed in mainly two ways: one is to apply process-driven methods and the other one is to apply data driven methods (Li et al., 2020b; Qiu et al., 2020). The process-driven method, which is based on the hydrodynamic laws, is a powerful tool to develop prediction models (Gelda et al., 2019; Han et al., 2021), however, process-driven modelling requires the detailed information on the tributaries and pollution sources, and also requires deep understanding on the physical and chemical processes (Garcia-Alba et al., 2019; Zhou, 2020). On the other hand, compared to process-driven modelling, data driven modelling methods only use the observed data to draw mathematical relationship between variables, and therefore it does not require complex theories and processes such as model calibration (Fijani et al., 2019; Wu et al., 2021). Furthermore, data driven models that are developed for one site can be easily applied to other sites (Zheng et al., 2021; Zhou, 2020). Therefore, in view of the advantage of data-driven method, it has been broadly studied by many scientists (Ngouna et al., 2020; Rousso et al., 2020). Several data-driven models such as autoregressive integrated moving average (ARIMA) (Hernandez et al., 2017), multiple linear regression (Rajaee and Boroumand, 2015), generalized autoregressive conditional

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Abbreviations

BiGRU	bidirectional gated recurrent unit
DO	dissolved oxygen
CEEMDAN	complete ensemble empirical mode decomposition with adaptive noise
D_{Stat}	direction change statistics
EWT	empirical wavelet transform
FCM	fuzzy C-means clustering
GRU	gated recurrent unit
LSTM	long short term memory
MAE	mean absolute error
MAPE	mean absolute percentage error
NH_3–H	ammonia nitrogen
RMSE	root mean squared error
SDE	standard deviation error
SVR	support vector regression
VMD	variational mode decomposition

heteroskedasticity (Wu et al., 2012) were used for water quality forecasting. However, these traditional statistics-based models could not consider the temporal order of input data, and so they are not proper for time series forecasting.

With the development of the artificial intelligence, machine learning models are becoming a focus for data driven modelling (Chen et al., 2020; Niu et al., 2020; Sit et al., 2020). Jimeno-Saez et al. (2020) used support vector regression (SVR) to predict Chlorophyll-a concentration of the Mar Menor (the largest hypersaline costal lagoon in Europe). Li et al. (2021a) proposed a hybrid model that applied long short term memory (LSTM) and convolutional neural network (CNN) for wastewater treatment system. Zhi et al. (2021) applied LSTM to develop forecast model for dissolved oxygen. Miao et al. (2021) compared the forecast performance of SVR, LSTM, and gated recurrent unit (GRU) for chemical oxygen demand (COD) forecasting. In the researches described above, machine learning models, particularly, the deep learning models such as LSTM and GRU, showed great capacity for water quality forecasting.

However, in some cases, single machine learning models could not provide satisfied forecast result because water quality generally varies in a very complicated way, and therefore, some researches proposed several types of hybrid models, such as data assimilation models (Wang et al., 2016) and theory guided models (Karpatne et al., 2017).

These models, however, directly uses single scale data and so they can only use the surface features of the data. Studies showed that the models based on the multi scale data, that is to say, the models based on data decomposition can extract more detailed information from the original time series so that more accurate prediction can be obtained by this kind of models (Eze and Ajmal, 2020; Rajaee and Boroumand, 2015). Making the use of the data decomposition in developing forecast models, many researchers developed new type of hybrid models that combined data decomposition and machine learning algorithms: wavelet decomposition-based forecast models (Xiao et al., 2017; Zhang et al., 2018), empirical mode decomposition-based forecast models (Cao et al., 2019; Liang et al., 2019; Lu and Ma, 2020), variational mode decomposition-based forecast models (Huang et al., 2021; Pipelzadeh and Mastouri, 2021), empirical wavelet transform-based forecast models (Liu et al., 2021a), etc. From the previous research results, it can be seen that the data decomposition skill is an effective method to enhance simulation capacity of a regression model.

However, most of the studies on data decomposition-based forecast model applied one-time data decomposition method, which uses the decomposed result of the whole data for either model training or testing. Since the future values that are supposed to be known are used in the decomposition process, this method is not of practical use (Liu et al., 2020). In order to solve this problem, several recent studies suggested real-time data decomposition-based models (Kim et al., 2021). Some researches, however, stated that real-time data decomposition-based forecast models provided worse result than single models that did not combine data decomposition. This is because in real-time decomposition, the decomposed result obtained by step-by-step decomposition is of different trend from each other, and so it extracts different relationship at each step to result in failure in regression modelling (Ahani et al., 2020; Wang and Wu, 2016). In other words, for a successful application of data decomposition to develop forecast model, it is essential to design a proper data decomposition strategy and modelling method. Water quality time series data is of different characteristic from other time series data, such as air quality or wind speed. It differs from others in many aspects such as monitoring period and temporal correlation, etc (Rashed and Younis, 2012; Snortheim et al., 2017). Therefore, it is practically significant to develop a real-time data decomposition-based model for water quality forecasting.

In this paper, we propose a new data decomposition-based prediction model by using empirical wavelet transform (EWT), fuzzy C-means clustering (FCM) and bidirectional gated recurrent unit (BiGRU).

The main novelties of this research are as follows.

1) FCM is employed to reduce the unnecessary errors caused by real-time data decomposition. In a data decomposition-based forecast model, the large number of modes usually causes negative impact for prediction accuracy (Sun and Li, 2020; Wu and Lin, 2019).
2) BiGRU is applied to water quality forecasting for the first time. GRU only considers one-way relationship, and so it could not fully represent the relationship for the data with bidirectional correlation (Ahmed and Ghabayen, 2020; Meng et al., 2021). Since BiGRU consists of two GRUs (forward-propagated and backward-propagated) to consider the bidirectional relationship, BiGRU can be expected as a better option for water quality forecasting than GRU.
3) A new data decomposition-based modelling method is proposed. Differently from the existing data decomposition-based modelling methods (Ahani et al., 2020; Wang and Wu, 2016), the proposed method collects the decomposed result at each time point to consider the overall dynamic trend and construct the input-output relationship.

2. Study site and water quality data

2.1. Study site

Poyang Lake (Fig. S1) is the largest freshwater lake in China, located in the northern part of JiangXi Province (115°47′-116°45′E, 28°22′-29°45′N). It is a typical throughput lake which is wide and shallow in south, narrow and deep in north (Zhang et al., 2016). The water from Gan River, Rao River, Fu River, Xiu River, and Xin River flows into Poyang Lake, and it again flows into Yangtze River (Li et al., 2021b; Ma et al., 2021). The areas around Poyang Lake is of warm and humid subtropical climate, with average annual temperature of 17.3 °C and average annual rainfall of 1358–1823 mm (Liu et al., 2021b). Due to the seasonal changes, water level of the lake fluctuates from 8 m to 16 m and the surface area reaches 3000 km^2 in monsoon seasons, whereas it reaches 500 km^2 in dry seasons (Zhang et al., 2021).

Poyang Lake is an important ecological area included in The List of Wetlands of International Importance of “Ramsar Convention” (Ramsar, 2004), which performs various functions such as drinking water supply, irrigation, climate regulation and biodiversity maintenance (Li et al., 2020a). Runoff from Poyang Lake accounts for about 18% of the volume of Yangtze River which supplies drinking water to 40% of the Chinese population (Wang et al., 2020). Until 2003, the water quality of Poyang Lake reached the level of Class III according to the “Environmental

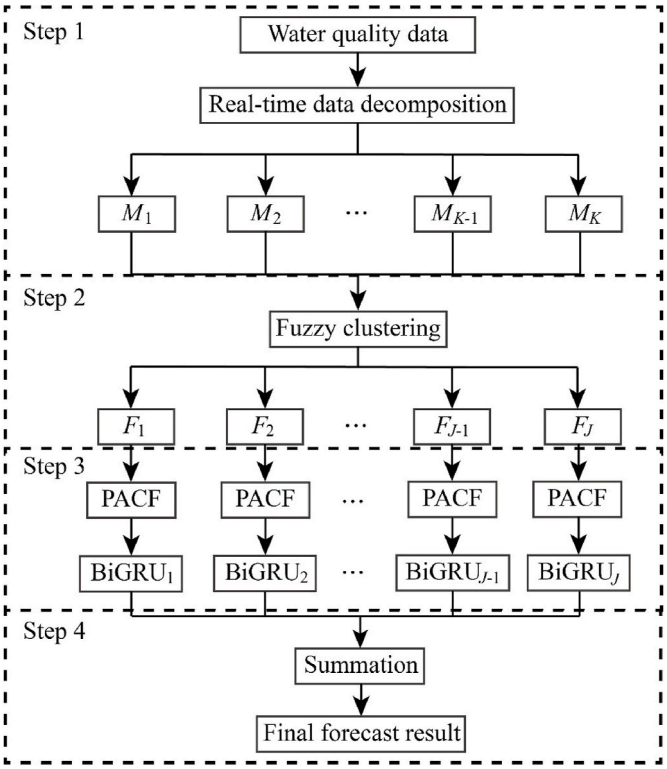


Fig. 1. Flowchart of the proposed forecast method.

Quality Standard for Surface Water: GB 3838–2002″, that is, it satisfied the standard of secondary drinking water reservoir and fishing ground (Wang et al., 2015). However, in recent years, the increase of urbanization, industrialization in the surrounding areas caused great quantity of various pollutants continue to flow into the lake, and consequently, resulted in contamination of lake water (Wang et al., 2020). In view of the importance of the serious impact of the water quality of Poyang Lake on ecological safety and socio-economic development of the surrounding areas, the researches on water quality of Poyang Lake have always been a focus. Therefore, in this study, we establish and verify a forecast model for water quality of Poyang Lake.

2.2. Data description

In this study, we use the water quality data of 3 monitoring stations (S_1, S_2, S_3) of Poyang Lake (Fig. S1). S_1 is located at the junction of Gan River and Poyang Lake, which reflects the load level of pollutants flowing from outside. S_2 is located at the outlet of Poyang Lake into Yangtze River, which reflects the influence of Poyang Lake on the Yangtze River. S_3 is located in the middle of the lake, which reflects pollution level of the water body of Poyang Lake itself. The water quality data is obtained from the website of the Ministry of Ecology and Environmental Protection of the People's Republic of China (https://www.mee.gov.cn).

Water quality data from each station consists of daily observed data of 2 water quality parameters (dissolved oxygen (DO), ammonia nitrogen (NH_3–H)) from October 1, 2017 to April 30, 2020. The dataset of the first 15 months is used for model training, the dataset of the next 2 months is used for model validation, and the remaining data of 2 months is used for model testing. Fig. S2 graphically shows the water quality data, and Table S1 presents the statistical information. As can be seen from Fig. S2, water quality of Poyang Lake obviously changes from season to season, and it is clear that all the values in winter are considerably higher than in summer. In addition, from Table S1, it is evident the water quality of S_1 (lake entrance) is worst, and water quality of S_2 (lake outlet) is best. It shows that the pollution of the lake is mainly caused by the heavily polluted tributaries flowing from the west of the lake.

In the rest of this paper, we denote NH_3–H data of S_1–S_3 by D_1-D_3, DO

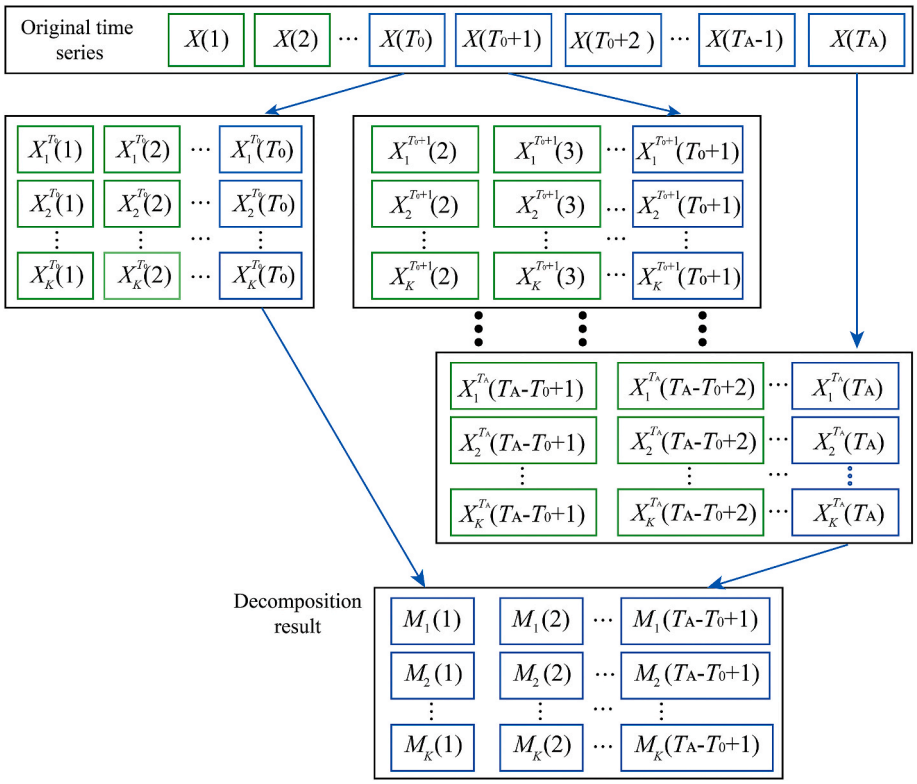


Fig. 2. Proposed data decomposition strategy.

Table 1
Forecast performance of the proposed model.

Data Type	MAE (mg/L)	RMSE (mg/L)	MAPE (%)	D_{stat} (%)	NSE
D_1	0.043	0.062	6.27	79.66	0.83
D_2	0.013	0.023	9.79	94.92	0.94
D_3	0.008	0.015	9.02	84.75	0.91
D_4	0.089	0.124	1.25	84.75	0.94
D_5	0.023	0.029	0.29	66.10	0.94
D_6	0.058	0.087	0.90	84.75	0.98
Average	0.039	0.057	4.59	82.49	0.92

data of S_1–S_3 by D_4-D_6.

2.3. Temporal correlation analysis

Temporal correlation analysis on the water quality data is conducted (Fig. S3). From Fig. S3, we can see that temporal correlation of NH_3–H (D_1-D_3) is much lower than DO (D_4-D_6). For a time series with low temporal correlation such as D_1-D_3, it is not usually expected to obtain highly accurate prediction result by a single statistical model (Li et al., 2020b; Xu et al., 2020). One of our main goals, therefore, is to develop a forecast model that can provide good result for the data of low temporal correlation.

3. Methodology

3.1. Overview of forecasting process

The forecast process of our model is presented below (Fig. 1).

Step 1. Water quality time series is decomposed into several components (see 3.2).
Step 2. The decomposed components are recombined by FCM (see 3.3).
Step 3. For each subseries obtained in Step 2, partial autocorrelation function (PACF) analysis is conducted to get proper input data from its time lagged subseries, and then, BiGRU is employed to get prediction result (see 3.4).
Step 4. The final forecast result is obtained by the summation of the prediction results obtained in Step 3 (see 3.4).

3.2. Data decomposition

In order to decompose a data, first, we need to select an appropriate decomposition algorithm. In this study, we selected empirical wavelet transform (EWT) as our decomposition algorithm because it is an efficient one that has both the mathematical basis of wavelet transform (WT) and the adaptability of empirical mode decomposition (EMD) (Gilles, 2013). EWT was proved by some researches to be effective in real-time data decomposition-based modelling (Kim et al., 2021; Liu et al., 2021a). The detailed information on EWT is described in Appendix A.

The proposed data decomposition strategy is as follows (Fig. 2).

Let $\{X(t)\colon t = 1,2, \ldots,T_A\}$ is the whole time series data and $\{X(t)\colon t = 1,2, \cdots,T_T\}$ is the training dataset.

1) Select a proper number $T_0<T_T$, and decompose $\{X(t)\colon t = 1, 2, \cdots, T_0\}$ into $\{X_i^{T_0}(t) : t = 1, 2, \cdots, T_0\}$ $(i = 1, 2, \cdots, K)$ by EWT (K: number of decomposed subseries). Here, we only select the decomposed result of the end point $X(T_0)$, and denote $X_i^{T_0}(T_0)$ by $M_i(1)$ (i=1,2,⋯,K).

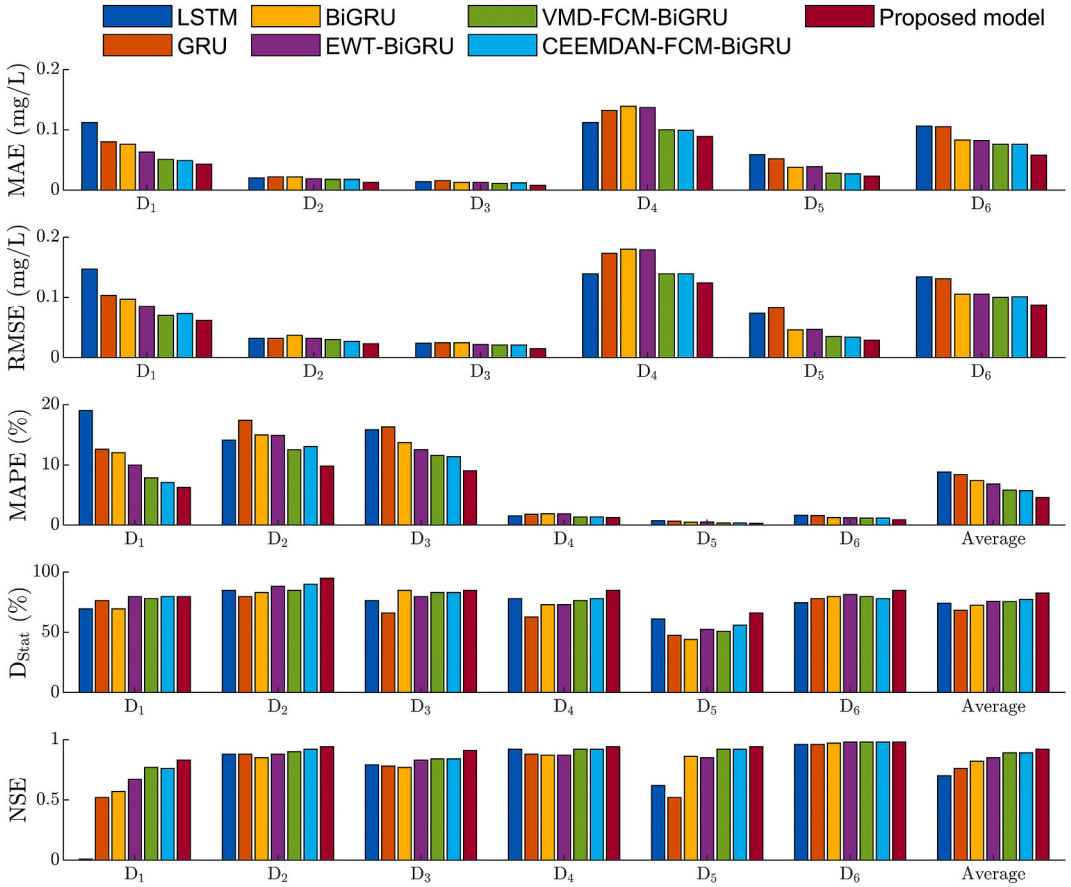


Fig. 3. Comparison of forecasting performance (1).

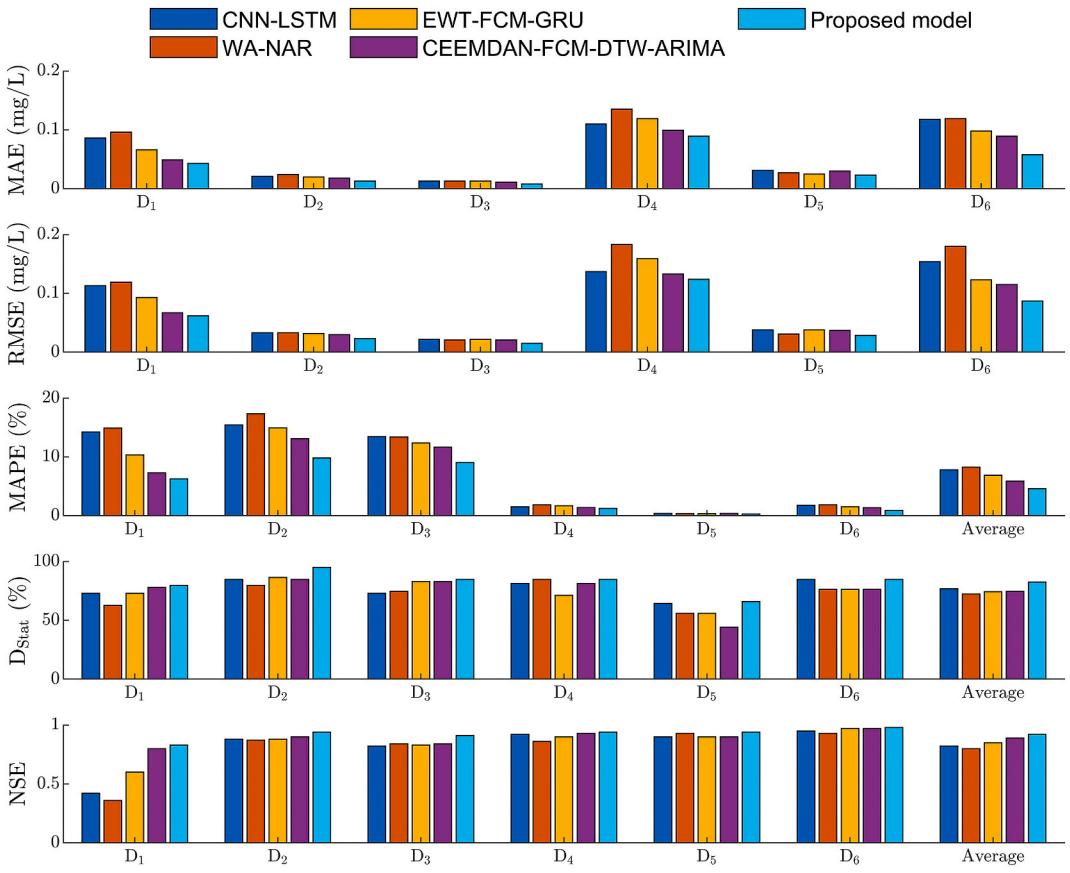


Fig. 4. Comparison of forecasting performance (2).

2) For each $T \in \{T_0 + 1, T_0 + 2, \cdots, T_A\}$, decompose $\{X(t): t = T - T_0 + 1, T - T_0 + 2, \cdots, T\}$ into $\{X_i^T(t) : t = T - T_0 + 1, T - T_0 + 2, \cdots, T\}$ $(i = 1, 2, \cdots, K)$ by EWT, and select the decomposed result of the end point $X(T)$, denote $X_i^T(T)$by $M_i(T\text{-}T_0+1)$ $(i=1,2,\cdots,K)$.

Finally, the decomposed result $\{M_i(t) : t = 1, 2, \cdots, T_A - T_0 + 1\}$ $(i = 1, 2, \cdots, K)$ is obtained.

3.3. Recombination by fuzzy C-means clustering

For an application of fuzzy C-means clustering (FCM), it is needed to determine the distance between the components to be clustered. In this study, dynamic time warping (DTW) is employed for this purpose because this distance not only can consider the scale difference between the series and also can take the time delay into account (Chai et al., 2019; Kim et al., 2021). The details of DTW and FCM are described in Appendix A.

Below we present the recombination process of the K subseries $\{M_i\}$ $(i = 1, 2, \cdots, K)$ obtained in 3.2 by using FCM based on DTW.

First, the following partition matrix is obtained by FCM.

$$U = [u_{ij}](i = 1, 2, \cdots, K; j = 1, 2, \cdots, J),$$

where $u_{ij} \in [0, 1]$, $\sum_{j=1}^{J} u_{ij} = 1$ $(i = 1, 2, \cdots, K)$, $0 < \sum_{i=1}^{K} u_{ij} < K$ $(j = 1, 2, \cdots, J)$, J is the number of clusters. Next, for M_i $(i = 1, 2, \cdots, K)$, we first find u_{ij} which is the maximum value of $\{u_{ik} : k = 1, 2, \cdots, J\}$ and put M_i in j^{th} cluster C_j. In this way, we categorize $\{M_i | i = 1, 2, \cdots, K\}$ into J clusters, and obtain new J subseries by summing the components of each cluster. Consequently, we now obtain J series $\{F_j | j = 1, 2, \cdots, J\}$.

3.4. Obtaining prediction by BiGRU

Many types of machine learning methods were applied to water quality forecasting in the previous researches. Among them, LSTM and GRU are drawing attention of the researchers for their long-term memory ability. Especially, GRU was widely applied in many studies due to its simple structure and high simulation capacity. However, since GRU only extracts the information in one direction, it could not sufficiently represent the bidirectional temporal correlation. BiGRU, which combines two GRUs, can successfully overcome the shortcoming of GRU because it extracts information in two directions by using backward GRU and forward GRU. BiGRU was already applied in several researches on wind speed and air quality forecasting, and was proved to be more effective than GRU. The details on BiGRU are presented in Appendix A.

Below we present a brief explanation about the process to obtain forecast result for original water quality data.

For each series F_j $(j = 1, 2, \cdots, J)$, input data is obtained from its time delayed data by taking the PACF values into account (see 4.1.2). From this input data and BiGRU, the prediction result for F_j is obtained. Let the prediction for F_j is F_j' $(j = 1, 2, \cdots, J)$. The final prediction result X' is obtained by the summation of $\{F_j'\}$ $(j = 1, 2, \cdots, J)$.

$$X' = \sum_{j=1}^{J} F_j'$$

3.5. Evaluation metrics

In this study, mean absolute error (MAE), root mean squared error (RMSE), mean absolute percentage error (MAPE), direction change statistics (D_{Stat}), Nash–Sutcliffe efficiency (NSE) are used to evaluate the forecast performance.

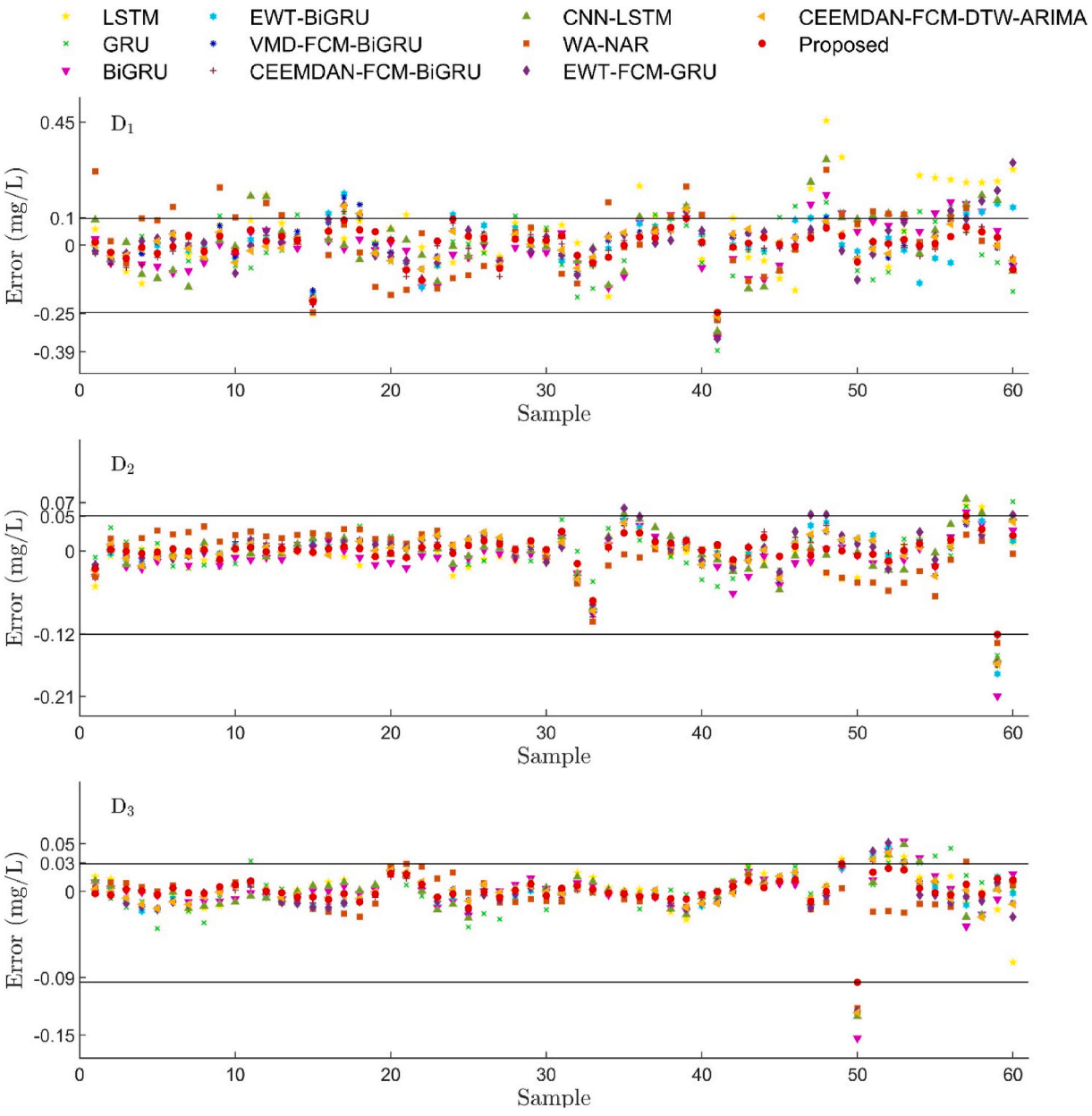


Fig. 5. Forecast errors.

Table 2
Standard deviation of error.

Model	SDE (mg/L)		
	D_1	D_2	D_3
LSTM	0.143	0.032	0.024
GRU	0.103	0.032	0.025
BiGRU	0.097	0.035	0.025
EWT-BiGRU	0.085	0.032	0.022
VMD-FCM-BiGRU	0.071	0.030	0.021
CEEMDAN-FCM-BiGRU	0.073	0.027	0.021
CNN-LSTM	0.113	0.033	0.023
WA-NAR	0.119	0.033	0.021
EWT-FCM-GRU	0.093	0.032	0.021
CEEMDAN-FCM-DTW-ARIMA	0.067	0.03	0.021
EWT-FCM-BiGRU	0.062	0.023	0.016

$$\mathrm{MAE} = \frac{\sum_{t=1}^{L}|O_t - P_t|}{L},$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{L}\sum_{t=1}^{L}(O_t - P_t)^2},$$

$$\mathrm{MAPE} = \frac{1}{L}\sum_{t=1}^{L}\frac{|O_t - P_t|}{O_t},$$

$$\mathrm{D_{Stat}} = \frac{1}{L}\sum_{t=1}^{L}\alpha_t, \alpha_t = \begin{cases} 1, & if\ (O_t - O_{t-1}) \times (P_t - O_{t-1}) \geq 0, \\ 0, & otherwise, \end{cases}$$

$$\mathrm{NSE} = 1 - \frac{\sum_{t=1}^{L}(O_t - P_t)^2}{\sum_{t=1}^{L}(O_t - \overline{O})^2}$$

where O_t is actual value, $\overline{O} = \frac{\sum_{t=1}^{L} O_t}{L}$, P_t is the prediction value, L is the

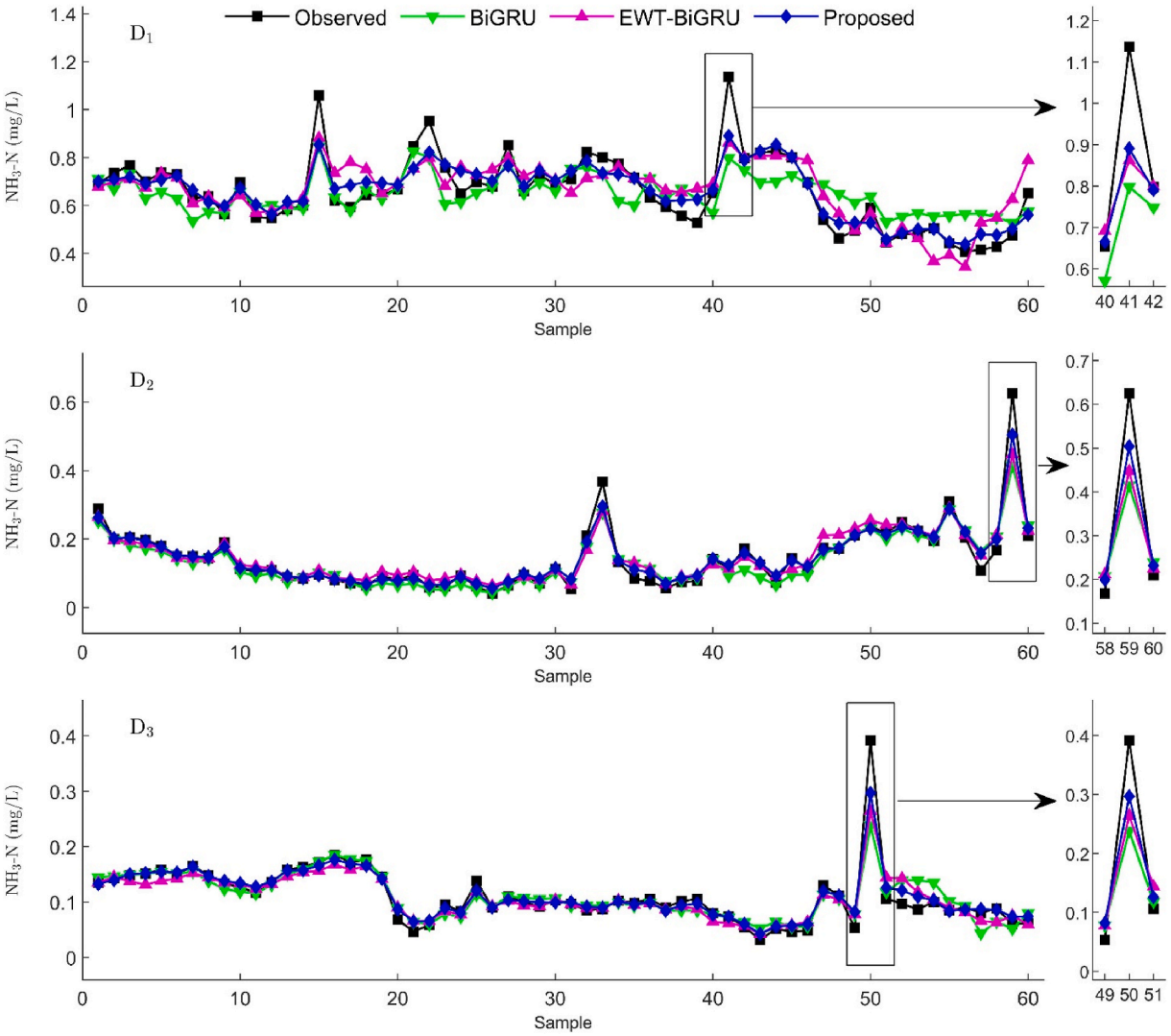


Fig. 6. Prediction result of BiGRU, EWT-BiGRU and EWT-FCM-BiGRU.

length of data.

These are commonly used metrics to assess the performance of a prediction model. Note that for MAE, RMSE and MAPE, the smaller value means the better prediction, while for D_{Stat} and NSE, the larger value means the better prediction.

4. Model construction and forecast result

4.1. Model construction

We evaluate the forecast capacity of our model through 7 day lead forecast. In this section, we describe the optimization process of our model. All the processes are implemented by Matlab (2020b).

4.1.1. Data decomposition

Taking into account the length of the total time series, we set the fixed length of the data to be decomposed to 200. In the decomposition process by EWT, the number of modes is automatically determined by identifying local peaks in a multi-taper power spectral estimate. Consequently, D_1-D_6 were decomposed into 7, 6, 6, 6, 7, 7 subseries, respectively. As can be seen from the decomposed result of D_1 (Fig. S4), the complexity of decomposed subseries are decreased than the original time series. Each decomposed subseries represents the components of different frequency involved in the original data: the component of high frequency (M_1) represents the detailed dynamic trend of original data, and the component of low frequency (M_7) represents the overall dynamic trend of original time series. Since we can acquire more detailed information by decomposing the original time series, we can expect to get more accurate result by using data decomposition rather than using the original time series.

4.1.2. Clustered series

The number of clusters is an important factor in our method because different number of clusters determines different structure of the forecast model and further, results in different forecast. In this study, the number of clusters were examined from 2 to 5, and we selected the optimal number that provided the most accurate prediction result. Table S2 shows the MAPE values of forecast result by different number of clusters. From Table S2, it is evident that MAPE is the smallest in case of 3. As the number of clusters increased from 2 to 3, MAPE decreased, and it shows that our model can extract more detailed relationship when the number of clusters is 3. On the contrary, MAPE increases as the number of clusters increased from 3, and it shows that the dynamic trend of the clustered series becomes more complicated as the number of clusters increases. From this experiment, we can draw that the optimal number of clusters for D_1 is 3. Fig. S5 shows the result when the number of clusters is 3. As Fig. S5 shows, M_1 and M_7 becomes an individual cluster F_1 and F_3, respectively, and the rest subseries (M_2-M_6) belong to

another cluster F_2. As a result, after clustering, we obtained 3 series F_1 (high frequency), F_2 (middle frequency), F_3 (low frequency).

4.1.3. Input selection

After the clustering is done, we conducted PACF analysis on the series to determine a proper time lag to select input data. For each subseries, we calculated PACF values and found the maximum time lag (τ) included in the 95% confidence interval (Table S3). And then, the size of input is set to $\tau - 6$, and at time point t, $\{F_i(t-\tau+7), F_i(t-\tau+6), \cdots, F_i(t)\}$ is set to the input data for time point $t+7$. For example, the determined maximum time lag for F_1 of D_1 is 23 and so, $\{F_1(t-15), F_1(t-14), \cdots, F_1(t)\}$ is used to as the input to predict $F_1(t+7)$ (Fig. S6).

4.1.4. Hyperparameters of BiGRU

The training and validation of BiGRU was performed by the input-output obtained in 4.1.3. Here, the hyperparameters of BiGRU were found out by trial and error. The optimized hyperparameters were shown in Table S4.

4.2. Forecast result

After the hyperparameters were optimized, the model is again trained by the sumset of the training data and validation data. And then, this model was used to predict the testing data. Table 1 shows the statistical information of the forecast result. From Table 1, we can see that the forecast for DO (D_4-D_6) is very accurate (MAPE <1.3%) and for NH_3–N, the proposed model also showed good forecasting performance (MAPE<10%). Although the forecast for NH_3–H is much worse than DO, this result can be considered as positive, taking into account the low temporal correlation of the original NH_3–H time series.

4.3. Comparison with other models

For a more thorough evaluation of the proposed forecast model, several other prediction models were developed and compared with our model. Our model is denoted by EWT-FCM-BiGRU and other models are denoted in a similar way. We implemented two experiments: in Experiment I, we compare our model with models with similar structure and in Experiment II, we compare with the models from the previous studies.

4.3.1. Experiment I

The structure of the forecast models is presented in Table S5. These models were developed in a similar way as EWT-FCM-BiGRU (see 4.1) and used to forecast the testing data. Fig. 3 shows the forecast by these models.

From Fig. 3, we can draw the following conclusions.

1) Among the models that did not apply data decomposition (LSTM, GRU, BIGRU), BiGRU provided the highest accuracy, and this shows the high simulation capacity of BiGRU.
2) Compared with the single models, the data decomposition-based models provided better forecast result. And this result shows that data decomposition is an effective method to enhance forecast capacity.
3) EWT-FCM-BiGRU provided better forecast result than EWT-BiGRU. It shows that optimized recombination of the decomposed subseries can be more effective than using the decomposed subseries. Even though EWT-BiGRU provided better result for D_1 than BiGRU, the improvement is not so apparent. Moreover, for D_4, forecast result by EWT-BiGRU is worse than BiGRU. From this, we can see that with the data decomposition alone, it could not enhance forecast capacity noticeably, and therefore, our modelling method that combines data decomposition and clustering method is significant.
4) Our model EWT-FCM-BiGRU showed better forecast result than VMD-FCM-BiGRU, CEEMDAN-FCM-BiGRU and this result proves that EWT is more suitable for developing forecast models than other decomposition algorithms.

4.3.2. Experiment II

In this part, 4 models proposed by previous researches (CNN-LSTM (Barzegar et al., 2020), WA-NAR (Zhang et al., 2018), EWT-FCM-GRU (Liu and Shen, 2020), CEEMDAN-FCM-DTW-ARIMA (Chai et al., 2019)) were compared with our model EWT-FCM-BiGRU. The forecast results by these models were shown in Fig. 4. As the result shows, our model outperformed all the other models with the best statical indicators.

5. Discussion

5.1. Stability of forecast

From the results described in 4.2, we can only assess the overall performance of a prediction model. However, in water quality forecasting, sometimes, it needs to get an accurate forecast for a specific time point. Additionally, model performance is also dependent on the data used. For example, for a data of high volatility, it is difficult to guarantee good forecast result. Therefore, in this regard, stability of a forecast is an important aspect to assess a prediction model. Considering the forecast for NH_3–N produced relatively high error, we use NH_3–N as an example to compare the stability of our model with other models. The forecast error by these models were shown in Fig. 5, and the corresponding SDE values were presented in Table 2. In Fig. 5, the black lines are the lower bound and upper bound of the forecast error by EWT-FCM-BiGRU. As can be seen from Fig. 5, some of the forecast errors by other models are distributed outside of the space between the two black lines. In addition, as Table 2 shows, SDE for our model is smaller than others. Overall, this result demonstrates our model can provide more stable and accurate forecast than other models.

5.2. Agreement between the forecasted and observed value

In this section, we graphically present the forecast result by BiGRU, EWT-BiGRU, EWT-FCM-BiGRU, so as to show the advantage of the proposed method more intuitively. As Fig. S7 shows, the forecast result by EWT-FCM-BiGRU is most concentrated around the regression line. In addition, as Fig. 6 shows, the forecast values by EWT-FCM-BiGRU is closer to the actual data than other models. It is more apparent for the extrema (41st point of D_1, 59th point of D_2, 50th point of D_3) and it shows that our model can provide a stable forecast even for a complex data.

6. Conclusion

In this study, we proposed a new water quality forecast model by combining EWT, FCM and BiGRU. In the proposed forecast model, EWT was applied to decompose the original water quality time series into several components and then, FCM was used to recombine the decomposed subseries into three components (low, middle and high frequency component). Subsequently, for each clustered component, BiGRU was employed to build prediction model. The proposed forecast model was evaluated by the water quality data of Poyang Lake, China.

Two kinds of experiments were conducted to assess the forecast performance of our model. The first experiment showed that our model is better than the models with similar architecture, and it proved the effectiveness of our modelling method, including selection of decomposition algorithm, application of FCM and BiGRU. The second experiment showed that our model could provide more accurate forecast than the models suggested by the previous researches, and it sufficiently showed that the advantage of our model. Moreover, forecast stability assessment showed that our model provided more stable prediction result than other models. The overall experiment results demonstrated

that the proposed model was very successful for water quality forecasting. It also should be noted that this study proposed a new forecast method that applied a novel real-time data decomposition strategy and BiGRU to water quality forecasting for the first time. Future work will be conducted to develop other type of forecast models in consideration of the trend of other factors.

Author contributions

Jin-Won Yu: Conceptualization, Methodology, Data collection and curation, Software, Investigation, Draft preparation. Ju-Song Kim: Draft preparation. Xia Li: Reviewing, Editing, Supervision. Yun-Chol Jong: Methodology, Data process, Visualization. Kwang-Hun Kim: Data process, Visualization. Gwang-Il Ryang: Data process, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.envpol.2022.119136.

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